

The Monetary System

Mbak 9515: Economics

Sungjun Huh

Konkuk University

Readings

- Chapter 11. The Monetary System
 - ▶ The three functions of Money
 - ▶ Money creation of commercial banks
 - ▶ The Fed's three ways to control money supply

Meaning of Money

- Money (or the “money supply”): any asset that can be used in making purchases
 - ▶ e.g.) Currency: paper bills and coins
 - ▶ Currency is too narrow to define money; consider checks and/or checking account
- Barter economy
 - ▶ Exchanging one good or service for another
 - ▶ Trade requires double coincidence of wants
 - ▶ Unlikely occurrence that two people each have a good or service that the other wants

Functions of Money

- **Medium of Exchange:**

- ▶ an asset used in purchasing goods and services
- ▶ Eliminates the trouble of finding a double coincidence of needs (reduces transaction costs)
- ▶ Promotes specialization: consider a barter economy

- A medium of exchange must:

- ▶ be easily standardized
- ▶ be widely accepted
- ▶ be divisible
- ▶ be easy to carry
- ▶ not deteriorate quickly

Functions of Money

● Unit of Account:

- ▶ a basic measure of economic value
- ▶ Reduces transaction costs
 - ★ e.g.) The economy with only three goods: apples, my lecture, and Avengers: Endgame

● Store of Value:

- ▶ an asset that serves as a means of holding wealth
- ▶ Other assets also serve this function
- ▶ Money is the most liquid of all assets but loses value during inflation.

Functions of Money

Examples

- **Credit Card**

- ▶ Satisfies a medium of exchange and a (negative) store of value
- ▶ **not** a unit of account
 - ★ a computer does not cost 10 VISA cards

- **A Paul Gauguin painting**

- ▶ a store of value only

- **Bitcoin**

- ▶ functions as a medium of exchange
- ▶ a unit of account and a store of value???
- ▶ <https://www.coindesk.com/price/>

Evolution of the Payments System

- Recall the definition of Money: anything that is generally accepted as **payment** for goods or services or in the repayment of debts.
- Commodity Money: valuable, easily standardized and divisible commodities
 - ▶ e.g.) gold, silver, **cigarettes**
 - ▶ *drawbacks*: difficult to carry around and hard to divide
- Fiat Money: paper money decreed by governments as legal tender
 - ▶ e.g.) paper currency and coins
 - ▶ *drawbacks*: easily stolen, difficult to track, and difficult to transport

Measuring Money

- How do we measure money? Which particular assets can be called “money”?
- Construct monetary aggregates using the concept of **liquidity**:
 - ▶ **M1** (most liquid assets) = currency + demand deposits + other checkable deposits + traveler's checks
 - ▶ **M2** (adds to M1 other assets that are not so liquid) = M1 + savings deposits and money market deposit accounts + small denomination time deposits + money market mutual fund shares

Measuring Money

[Note] Liquidity

- Investopedia defines **Liquidity** as the degree to which an asset or security can be quickly bought or sold in the market without affecting the asset's price
- People prefer liquidity:
 - ▶ This explains why people want to hold *Money*
 - ▶ e.g.) An extra \$1,300: you can either put it in your deposit account or purchase Google stock
 - ★ Deposit account → you will have \$1,300 in cash whenever you want to use
 - ★ Stock → you don't know how much cash you will have (price could soar or fall)

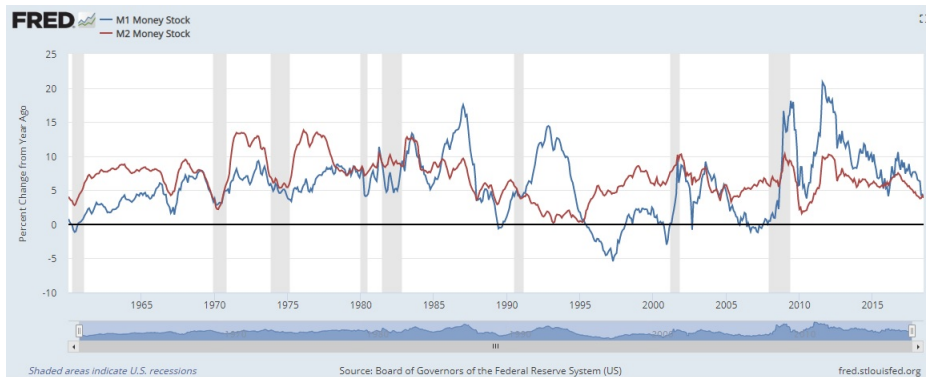
Measuring Money

TABLE 1 Measures of the Monetary Aggregates

	Value as of August 18, 2014, (\$ billions)
M1 = Currency	1,206.1
+ Traveler's checks	3.3
+ Demand deposits	1,089.9
+ Other checkable deposits	<u>477.4</u>
Total M1	2,776.7
M2 = M1	
+ Small-denomination time deposits	533.0
+ Savings deposits and money market deposit accounts	7,338.2
+ Money market mutual fund shares (retail)	<u>642.5</u>
Total M2	11,290.4

Source: <http://www.federalreserve.gov/releases/h6/hist>.

Growth Rates of the M1 and M2 Aggregates, 1960-2018



Bank Lending Creates Money

- Bank: a firm that specializes in brokering between savers and borrowers, allowing borrowers to go to one firm instead of 1000s
- Banks lend their excess reserves
- Cash lend by a bank returns to the banking system when that loan is spent by the borrower
- This creates a deposit in the name of the person who supplied goods and services to the borrower
- The supply of money in the form of bank deposits has increased

$$M = \frac{M^B}{rr}$$

Currency and Deposits

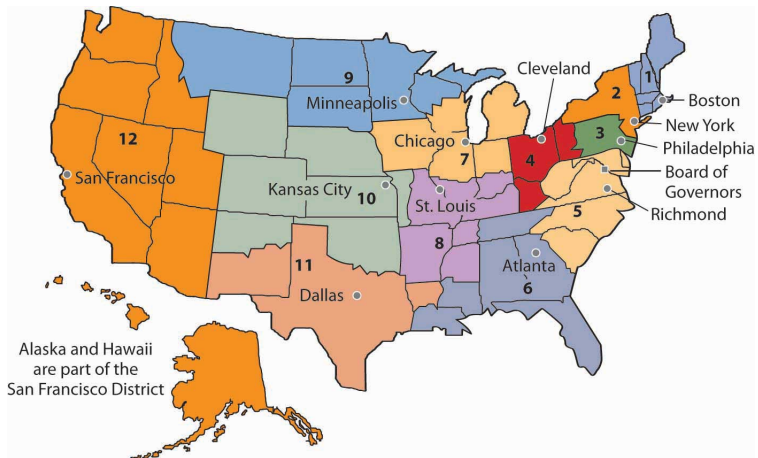
- Suppose that 1,000,000 dollars issued by the central bank and the citizens of Gorgonzola choose to hold a total of 500,000 dollars in the form of currency and to deposit the rest of their money in banks. Banks keep reserves equal to 10% of deposits. What is the money supply in Gorgonzola?

$$M = C + D$$

The Federal Reserve

- Federal Reserve System (the Fed)
 - ▶ The central bank of the United States
 - ▶ The Fed determines *how much money* circulates in the economy (Monetary policy)
 - ▶ Reduce *unemployment*, create *price* stability, and the smooth operation of *financial markets*
- After bank panics of the late 19th and early 20th century, the US created the Federal Reserve system in 1914
 - ▶ 12 regional Federal Reserve Banks
 - ▶ Board of Governors: the leadership of the Fed
 - ▶ Federal Open Market Committee (FOMC): the committee that makes decisions concerning monetary policy

The Federal Reserve



Money Supply

- **Monetary Policy:** managing the money supply in order to maximize employment and keep prices stable
- How does the Fed change the Money Supply?
 - ▶ Typically done through Open Market Operations: buying and selling of *Treasury bond*
- **Open market purchase** $\Rightarrow$ Federal Reserve purchases Treasury securities from banks and the public $\Rightarrow$ more money deposited in banks $\Rightarrow$ *Money Supply Increases!*
- **Open market sale** $\Rightarrow$ Federal Reserve sells Treasury securities that the government has created $\Rightarrow$ less money deposited in banks $\Rightarrow$ *Money Supply Decreases!*

Stabilizing Financial Markets

- The Fed also has the responsibility of the smooth operation of *financial markets*
- **Banking panic**: a situation in which news or rumors of the imminent bankruptcy of one or more banks leads bank depositors to rush to withdraw their funds
- The Fed was given the power to supervise and regulate banks
 - ▶ Greater confidence in banks
- The Fed was allowed to make direct loans to banks through a new facility called the discount window
 - ▶ Banks can borrow cash from the Fed
- cf) Federal Deposit Insurance Corporation (FDIC)
 - ▶ FDIC guarantees that depositors with deposits of less than \$250,000 will not lose money even if their bank goes bankrupt

In Sum...

Fed's Conventional Tools of Monetary Control

- Open Market Operations
- Reserve Requirement
- Discount Rate